1

$-\frac{2}{\pi}\cos(\pi xy)$

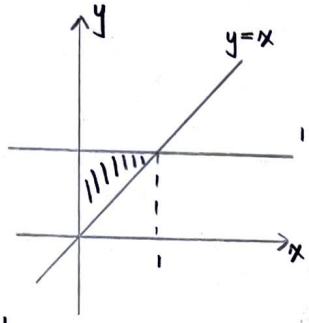
1. $\int_0^1 \int_0^1 2y \sin(\pi xy)\, dx dy$

$= \int_0^1 -\frac{2}{\pi}\cos \pi y + \frac{2}{\pi}\, dy$

$= \left[-\frac{2}{\pi^2}\sin \pi y + \frac{2}{\pi} y\right]_0^1$

$= \frac{2}{\pi}$ //

2. $\int_0^1 \int_x^1 \cos y^2\, dy dx$



$= \int_0^1 \int_0^y \cos y^2\, dx dy$

$= \int_0^1 [x \cos y^2]_0^y\, dy$

$= \int_0^1 y \cos y^2\, dy$

$= \int_0^1 \frac{1}{2}\cos t\, dt$

$= \frac{1}{2}$ //

3. $\int_0^1 \int_0^y \frac{1}{1+x^2}\, dx dy$ → $\tan^{-1} x$

$= \int_0^1 \tan^{-1} y\, dy$

* $\int \tan^{-1} y\, dy$

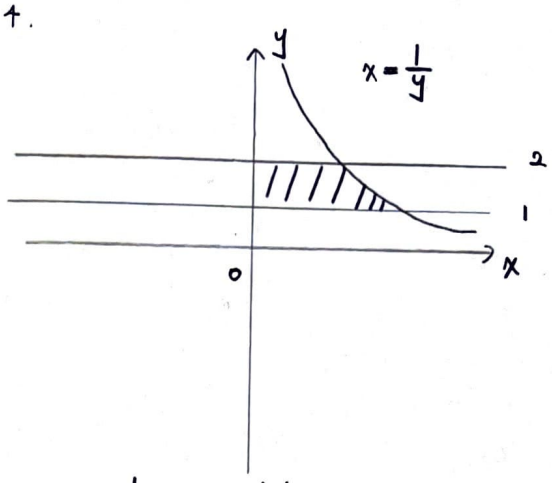
$= \tan^{-1} y \cdot y - \int \frac{y}{1+y^2}\, dy$

$= y \tan^{-1} y - \frac{1}{2}\ln(1+y^2) + C$

$\therefore \int_0^1 \tan^{-1} y\, dy$

$= \tan^{-1} 1 - \frac{1}{2}\ln 2$

4.

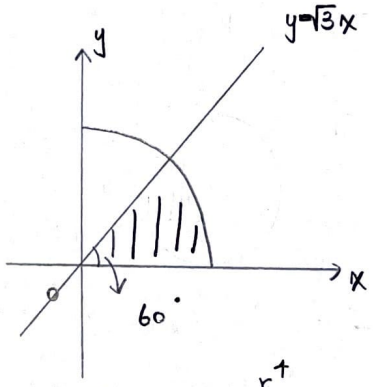


$\int_1^2 \int_0^{\frac{1}{y}} xy\, dx dy$ //

$= \int_1^2 1\, dy$

$= 1$ //

5.



$\theta = \frac{\pi}{3}$

$r = 3$

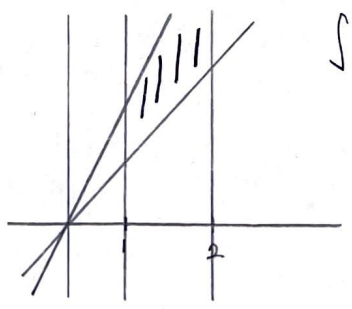
$\int_0^3 \int_0^{\frac{\pi}{3}} r^3 \cdot r\, d\theta dr$ // (r → r^4)

$= \int_0^3 \frac{\pi}{3} r^4\, dr$

$= \frac{\pi}{3} \cdot \frac{1}{5} \cdot 3^5$

$= \frac{81}{5}\pi$ //

6



$\int_1^2 \int_x^{2x} -\frac{6}{(x+y)^2}\, dy dx$ ($-\frac{6}{x+y}$ ↑)

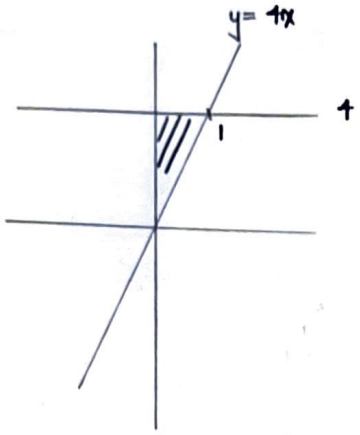
$\int_1^2 -\frac{2}{x} + \frac{3}{x}\, dx$

$= \int_1^2 \frac{1}{x}\, dx$

$= \ln 2$ //

7.

$$\int_0^1 \int_{4x}^{4} e^{-y^2}\, dy dx$$



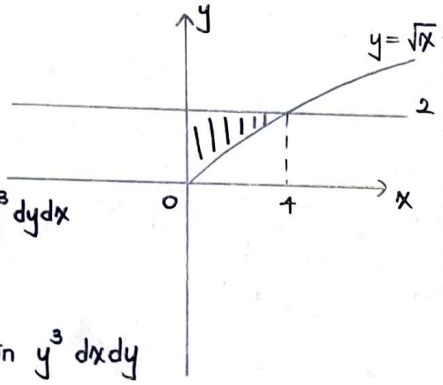
$$\therefore \int_0^4 \int_0^{\frac{y}{4}} e^{-y^2}\, dx dy$$

$$= \int_0^4 \frac{y}{4} e^{-y^2}\, dy$$

$$= \left[-\frac{1}{8} e^{-y^2} \right]_0^4$$

$$= -\frac{1}{8} e^{-16} + \frac{1}{8}$$

8.



$$\int_0^4 \int_{\sqrt{x}}^{2} \sin y^3\, dy dx$$

$$= \int_0^2 \int_0^{y^2} \sin y^3\, dx dy$$

$$= \int_0^2 y^2 \sin y^3\, dy$$

$$= \left[\frac{1}{3} \sin t \right]_0^2$$

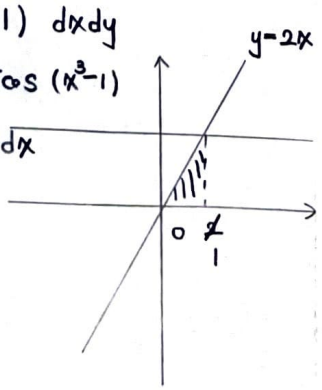
$$= \frac{1}{3} \sin 2 \quad //$$

9.

$$\int_0^2 \int_{\frac{y}{2}}^{1} y \cos(x^3-1)\, dx dy$$

$$\rightarrow \frac{1}{2}y^2 \cos(x^3-1)$$

$$= \int_0^1 \int_0^{2x} y \cos(x^3-1)\, dy dx$$



$$= \int_0^1 2x^2 \cos(x^3-1)\, dx$$

$$= \left[\frac{2}{3} \sin(x^3-1) \right]_0^1$$

$$= -\frac{2}{3} \sin(-1) \quad //$$

10.

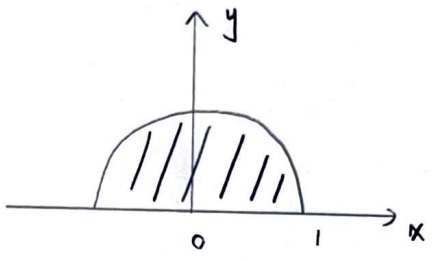
$$\int_0^1 \int_x^1 e^{\frac{x}{y}}\, dy dx$$

$$= \int_0^1 \int_0^y e^{\frac{x}{y}}\, dx dy \quad \rightarrow ye^{\frac{x}{y}}$$

$$= \int_0^1 ye - y\, dy$$

$$= (e-1) \cdot \frac{1}{2} \quad //$$

11.



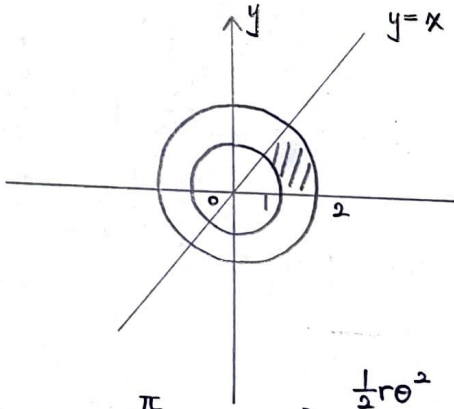
$$\int_0^1 \int_0^{\pi} e^{-r^2} \cdot r\, d\theta dr$$

$$= \int_0^1 \pi r e^{-r^2}\, dr$$

$$= \pi \cdot \left[-\frac{1}{2} e^{-r^2} \right]_0^1$$

$$= \pi \left(-\frac{1}{2} e^{-1} + \frac{1}{2} \right) \quad //$$

12.



$1 \le r \le 2$

$0 \le \theta \le \frac{\pi}{4}$

$$\int_1^2 \int_0^{\frac{\pi}{4}} \theta\, r \cdot d\theta dr \quad \rightarrow \frac{1}{2} r\theta^2$$

$$= \int_1^2 \frac{\pi^2}{32} r\, dr$$

$$= \frac{\pi^2}{32} \cdot \frac{1}{2} (4-1)$$

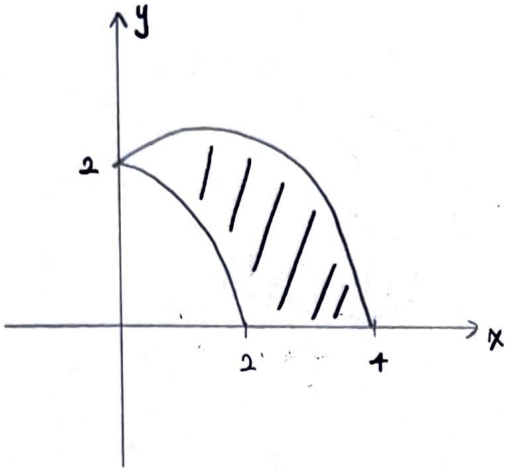
$$= \frac{3}{64} \pi^2 \quad //$$

13.

$x^2+y^2 = 2(1+x)$

$x^2-2x+y^2 = 2$ (+1)

$(x-1)^2+y^2 = 3$



$$\int_0^{\frac{\pi}{2}} \int_2^{2(1+\cos\theta)} r^2 \sin\theta \, dr\, d\theta \qquad \left(\frac{1}{3}r^3\right)$$

$$\int_0^{\frac{\pi}{2}} \sin\theta \cdot \frac{1}{3}\left((2(1+\cos\theta))^3 - 8\right) d\theta$$

$$= -\frac{2}{3} - \frac{8}{3} + \frac{32}{3}$$

$$= \frac{22}{3} \quad //$$

14.

$$\int_1^2 \int_0^1 \int_0^2 x^2yz \, dz\, dy\, dx \qquad \left(\frac{1}{2}x^2yz^2\right)$$

$$= \int_1^2 \int_0^1 2x^2y \, dy\, dx \qquad (x^2y^2)$$

$$= \int_1^2 x^2 \, dx$$

$$= \frac{1}{3}(2^3 - 1^3)$$

$$= \frac{7}{3} \quad //$$

15.

$$\int_{-2}^{5} \int_0^{3x} \int_y^{x+2} 1 \, dz\, dy\, dx \qquad (1z)$$

$$\int_{-2}^{5} \int_0^{3x} 1x+8-1y \, dy\, dx \qquad (1xy+8y-2y^2)$$

$$\int_{-2}^{5} 12x^2 + 24x - 18x^2 \, dx \qquad (-6x^2)$$

$$= \left[-2x^3 + 12x^2\right]_{-2}^{5} \qquad \begin{array}{r} 21 \\ 12 \\ \hline 21 \end{array}$$

$$= -2(125+8) + 12(25-4) \qquad (133)$$

$$= -266 + 252$$

$$= -14 \quad //$$

16.

$z = 2-x-2y$

$x = 2y$

$x = 0$

$z = 0$

$$\int_0^2 \int_0^1 \int_{\frac{x}{2}}^{1-\frac{x}{2}} 2-x-2y \, dy\, dx\, dz$$

$$= \frac{2}{3} \quad //$$